# Competing Risks and Multi-State Models

## Concepts, methods and software

Course given at *Karolinska Institute*

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# Part I

# Main Concepts and Quantities

Contents:

1. Examples
2. Intervention versus biological competing event
3. Main time-to-event quantities
4. Multi-state and subdistribution approach
5. Marginal or cause-specific distribution
6. Data structure

# Outline

## Examples Competing Risks

- All event types of interest
- One event type of interest, intervention as competing risk
- One event type of interest, biological event as competing risk

## On Rates and Risks

- Classical survival analysis
- Competing risks analysis
- Multi-state approach
- Subdistribution approach
- Type of estimands in the presence of competing risks

## Data Structure

- Time scales
- Right censored data
- Late entry/left truncation
- Setup data analysis

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## Event-type outcomes that are “competing” (mutually exclusive)

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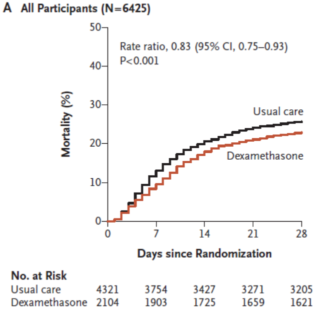
  - A: Effect of dexamethasone on recovery and death after SARS-CoV-2 infection (binary outcome, complimentary levels)
  - B: Spectrum in causes of death after HIV infection (role of cART)
  - C: AIDS and SI switch after HIV infection
  - D: Death and relapse after bone marrow transplant (Practical)

# A: Recovery (dexamethasone in Covid-19) trial




- Primary outcome: all-cause mortality within 28 days
- Secondary outcome: hospital discharge $\leq$ 28 days
- Binary outcome: death and recovery complimentary outcomes $\rightarrow$ logistic regression

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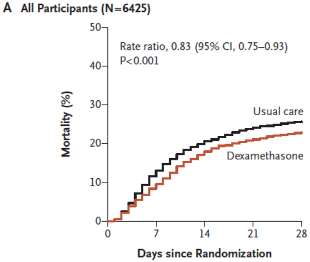


| No. at Risk | | | | | |
|---|---|---|---|---|---|
| Usual care | 4321 | 3754 | 3427 | 3271 | 3205 |
| Dexamethasone | 2104 | 1903 | 1725 | 1659 | 1621 |

- Primary outcome: all-cause mortality within 28 days
- Secondary outcome: hospital discharge $\leq$ 28 days
- Binary outcome: death and recovery complimentary outcomes $\rightarrow$ logistic regression
- But they used time-to-event methods
  - 10% event-free in hospital after 28 days event-free not a separate outcome
  - time to event may be of interest as well

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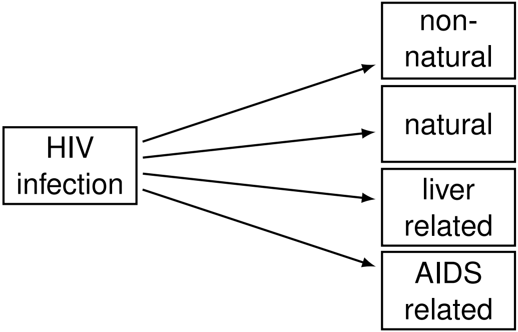
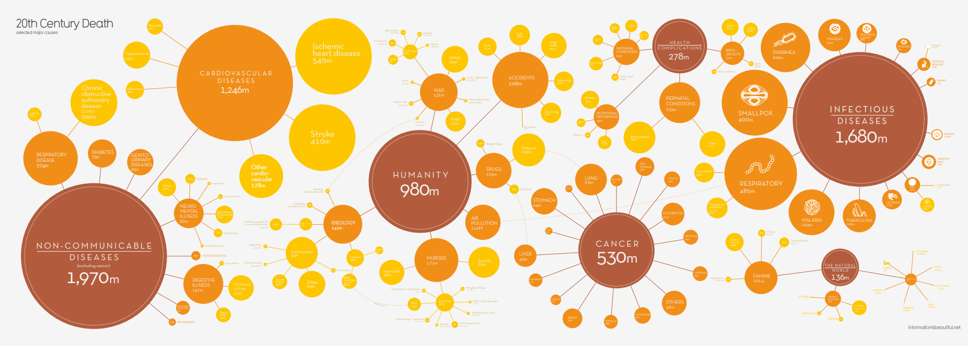
## B: Spectrum in causes of death after HIV infection

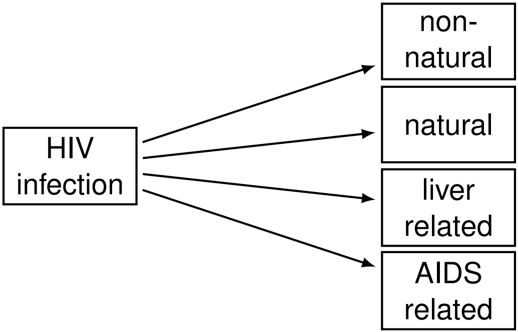



- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)



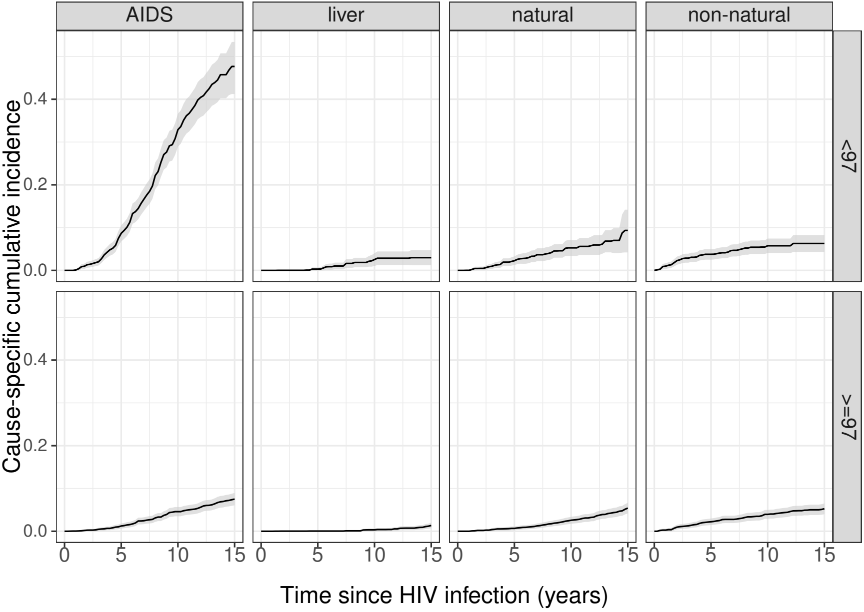

*In the end we all die, but not at the same time and not all from the same cause*

## B: Spectrum in causes of death after HIV infection




- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)
- Side effects of cART? *Etiology* ($\approx$ cause-specific hazard)
- In the end we all die. *Prediction* ($\approx$ subdistribution hazard)

Cause-specific mortality by calendar period



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# C: AIDS and SI switch after HIV infection

| |
|---|
| HIV infection; NSI phenotype |

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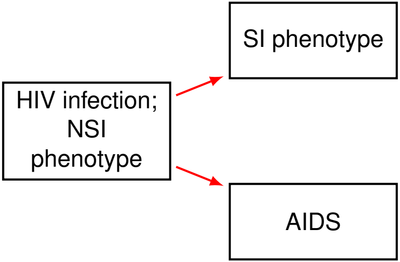


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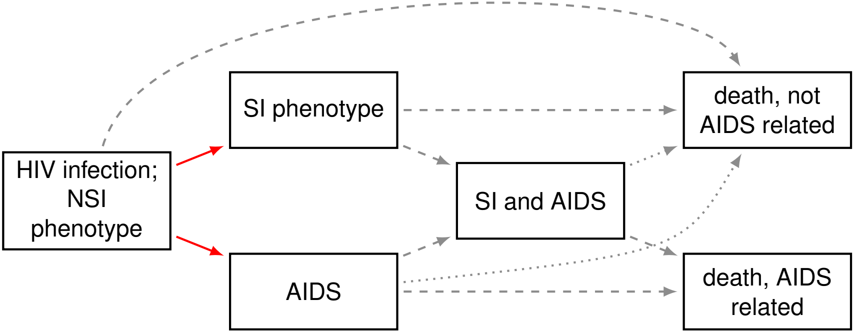


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- Multi state model also describes what happens after first event

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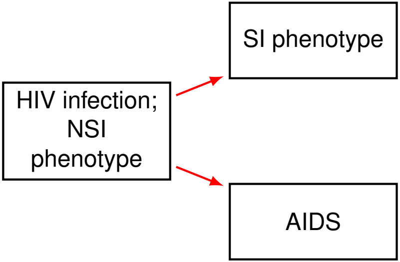


- NSI virus needs CCR5 co-receptor for cell entry
- Gene coding for CCR5 may have deletion CCR5-Δ32 on one chromosome
- Individuals with CCR5-Δ32 have slower AIDS progression

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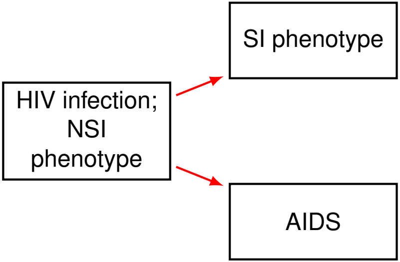


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- Question: SI switch as first event more likely for CCR5-Δ32 individuals?

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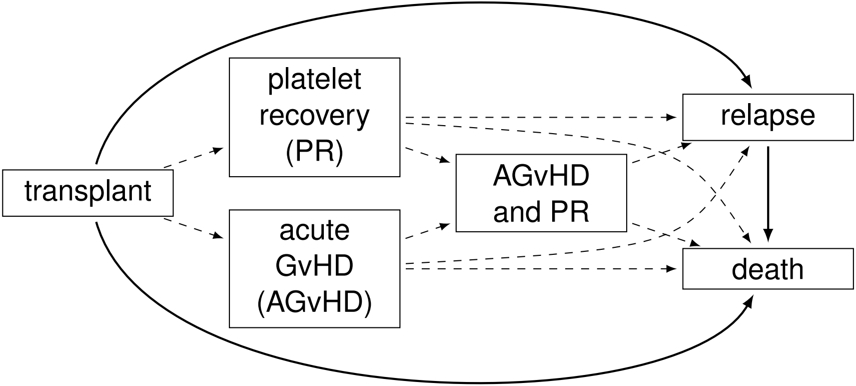
# D: Events after bone marrow transplantation



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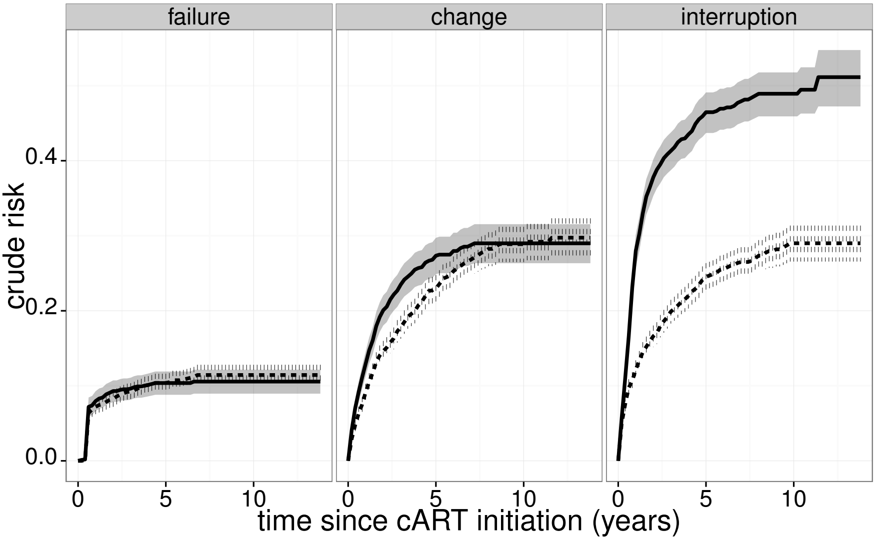
## F: cART treatment failure; early (solid) versus late (dashed) starters



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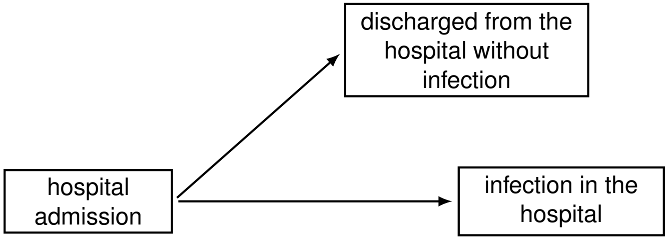
## G: Staphylococcus infection during hospital stay



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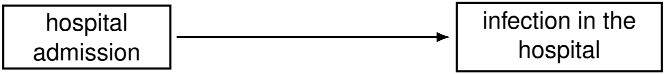


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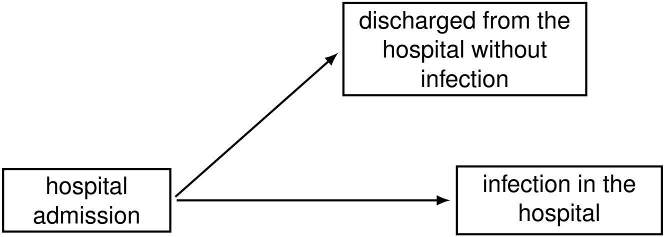


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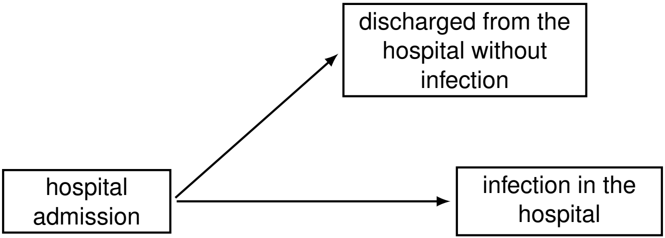


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# Estimation

| week | 0-1 | 1-2 | 2-3 | 3-4 | 4-5 | 5-6 | 6-7 | $> 7$ |
|---|---|---|---|---|---|---|---|---|
| infection | 1 | 2 | 6 | 11 | 9 | 11 | 2 | 18 |
| cumulative | 1 | 3 | 9 | 20 | 29 | 40 | 42 | 60 |
| | | | | | | | | |
| discharge | 5 | 9 | 6 | 6 | 9 | 12 | 4 | 35 |
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- Subdistribution. Estimated as frequency of events:

  $\widehat{P}(\text{infection} \leq 6\text{ weeks}) = 40/146$

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- Marginal distribution. Discharged individuals treated and interpreted as censored; Kaplan-Meier nonparametric estimator, assumes independent censoring

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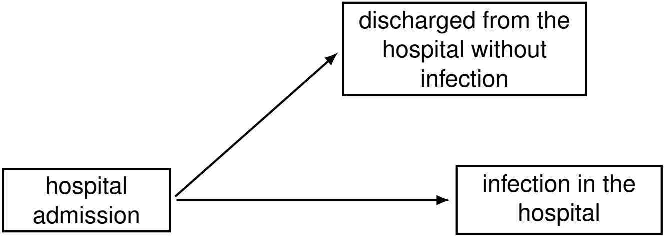


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- Which of two hospitals has higher risk may depend on type of risk

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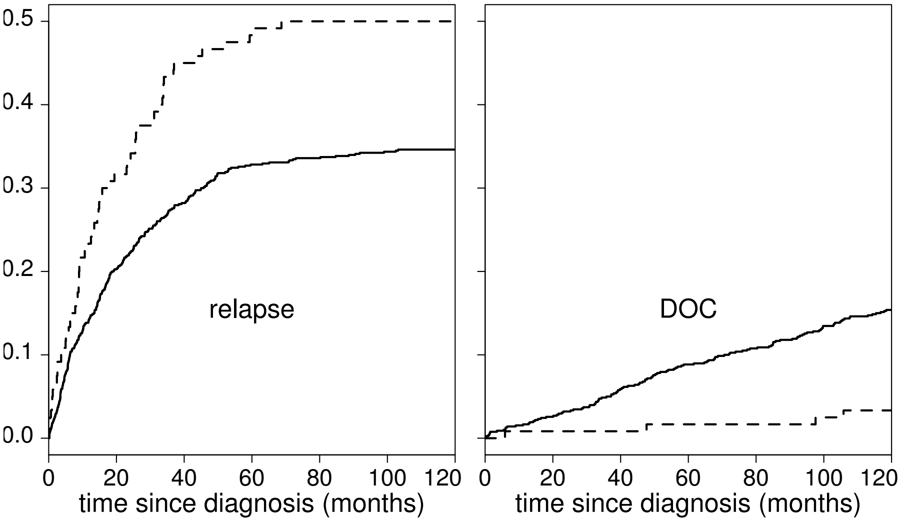
  - H: Death competing risk of non-fatal event (e.g. disease onset)

# H-1: Bladder cancer relapse, death other causes competing




Left: relapse. Solid: male; dashed: female

## H-2: Death prevents AIDS in HIV infected injecting drug users.

$$\boxed{\text{HIV}} \longrightarrow \boxed{\text{AIDS}}$$

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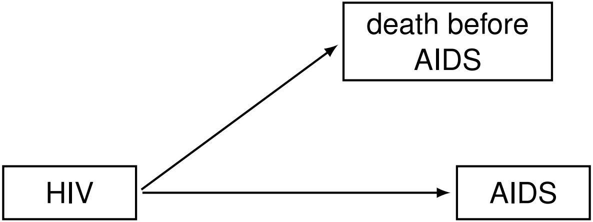


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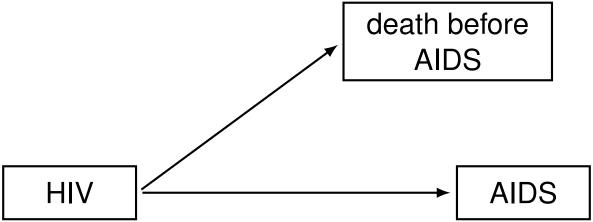


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- Compare men who have sex with men (MSM) and injecting drug users (IDU)
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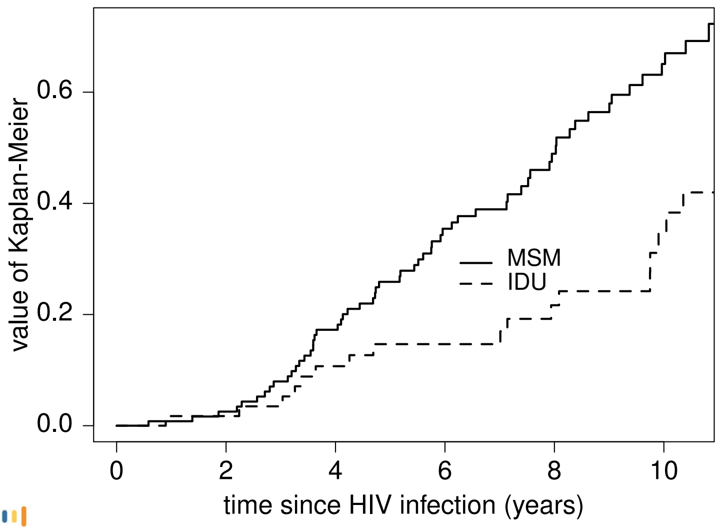
- Kaplan-Meier: leave risk set at death before AIDS
  Assumption: those that died had equal AIDS risk

# Kaplan-Meier: IDU much slower progression ($p = 0.001$)



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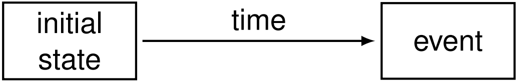

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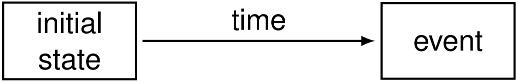


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  - Time. What is the time origin? Multiple time scales?

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# Main quantities in classical survival analysis

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- Risk/cumulative incidence $F(t) = \mathrm{P}(T \leq t)$
  survival function $\overline{F}(t) = 1 - F(t)$:
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Each of the three quantities completely describes distribution:

discrete Event times: $t_1, \ldots, t_L$

$$h(t_l) = \mathrm{P}(T = t_l \mid T \geq t_l)$$

(1)

continuous

(2)

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continuous

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \tag{2}$$

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$$= \mathrm{P}(T = t_l \mid T > t_{l-1}) = \frac{F(t_l) - F(t_{l-1})}{\overline{F}(t_{l-1})} \quad (1)$$

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$$\overline{F}(t) = \mathrm{P}(T > t) = \prod_{t_l \leq t} \{1 - h(t_l)\}$$

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$$\overline{F}(t) = \exp\left\{-\int_0^t h(s)ds\right\} = \exp\{-H(t)\}$$

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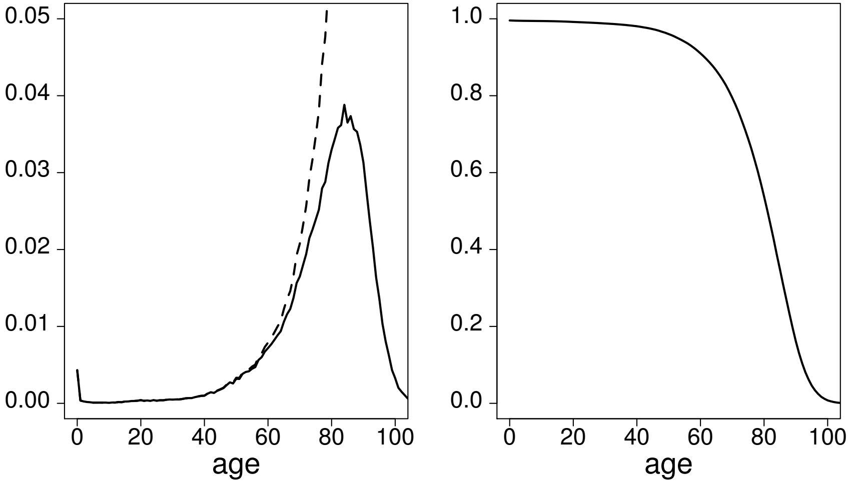

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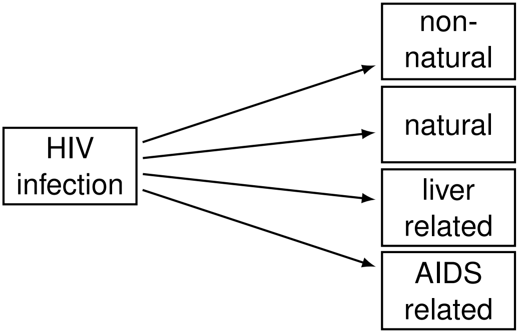
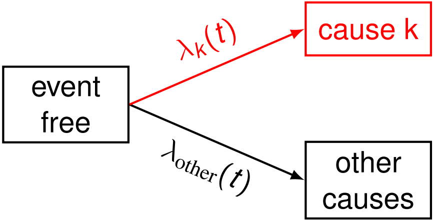
# B: Spectrum in causes of death after HIV infection




- Has the spectrum in causes of death changed after the introduction of cART (combination anti-retroviral therapy)
- Side effects of cART? *Etiology* ($\approx$ cause-specific hazard)
- In the end we all die. *Prediction* ($\approx$ subdistribution hazard)

# Outline

## The multi-state approach: cause-specific hazard




- Transition rate to cause k. For continuous distribution:

$$\lambda_k(t) = \lim_{\Delta t \downarrow 0} \frac{\mathrm{P}(t \le T < t + \Delta t, E = k \mid T \ge t)}{\Delta t}$$

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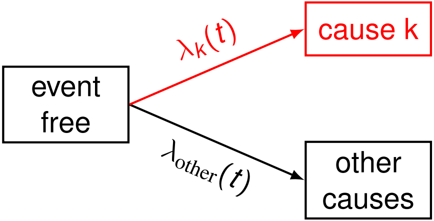


- Transition rate to cause k. For continuous distribution:

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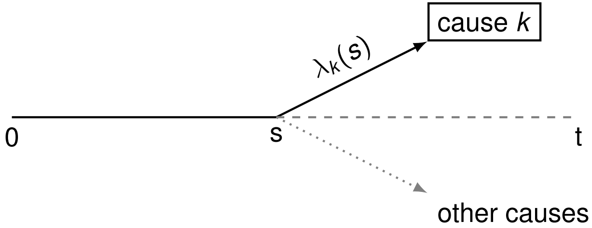
- Sum over causes is overall hazard: $\sum_{e=1}^{K} \lambda_e(t) = h(t)$
- Cause-specific hazard directly generalizes to multi-state setting (called transition hazard)

# From hazard to cumulative scale $\mathrm{P}(T \leq t, E = k)$

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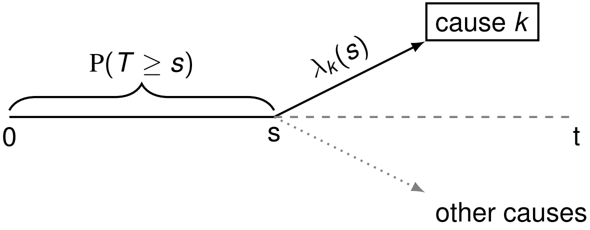
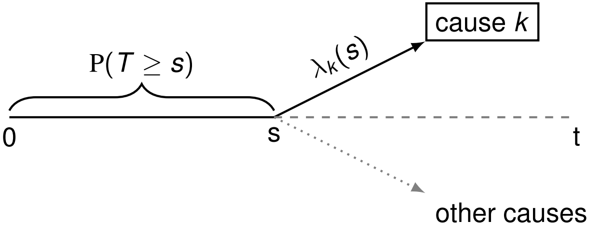



- $\overline{F}(s-)\lambda_k(s)$

## From hazard to cumulative scale $\mathrm{P}(T \leq t, E = k)$




- $F_k(t) = \mathrm{P}(T \leq t, E = k) = \int_0^t \overline{F}(s-)\lambda_k(s)ds$

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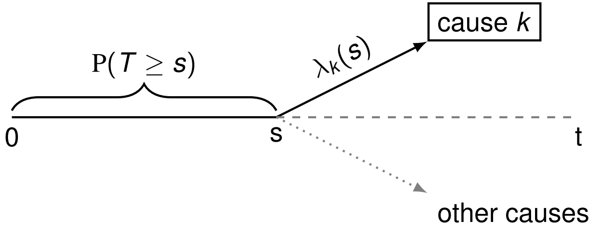


- $F_k(t) = \mathrm{P}(T \leq t, E = k) = \int_0^t \overline{F}(s-)\lambda_k(s)ds$
- Depends on all cause-specific hazards via overall "survival"

$$\overline{F}(s) = \exp\left\{-\int_0^s h(u)du\right\} = \exp\left\{-\sum_{e=1}^{K}\int_0^s \lambda_e(u)du\right\}$$

$\Longrightarrow \lambda_k$ does not uniquely specify $\mathrm{P}(T \leq t, E = k)$

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$$
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  - Not a rate in epidemiological sense, unless we can include immune individuals in the risk set (e.g. vaccination; unobserved cure)
- One-to-one relation with crude risk

$$\overline{F_k}(t) = \prod_{t_l \leq t} \left\{ 1 - h_k(t_l) \right\} \text{ or } \overline{F_k}(t) = \exp\{-\int_0^t h_k(s) ds\}$$

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# Rates and risks in competing risks setting

| | hazard | | cumulative | |
|---|---|---|---|---|
| competing risks | marginal | $\lambda^m$ | net risk<br>marginal survival function<br>marginal cumulative incidence | $F_k^m(t)$ |
| | cause-specific | $\lambda_k$ | no corresponding quantity | |
| | subdistribution | $h_k$ | crude risk<br>cause-specific cumulative incidence | $F_k(t)$ |
| combined | overall | $h$ | overall risk<br>overall survival function<br>overall cumulative incidence | $F(t)$ |

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# Exercise I.5.2

Psychiatric patients are treated in two different hospitals, A and B. We want to investigate whether the effect of a treatment on the development of side effects is the same for both hospitals. Hospital A collects information on side effects after discharge, and hospital B does not. In our analysis for hospital A we also use information after discharge, while for hospital B we censor individuals when they are discharged.
Is discharge a competing risk? What type of distribution do we want to quantify? Under which conditions can this be a fair comparison between both hospitals?

# Exercise I.5.3

We want to compare the cancer event rates in the virtual situation when the competing risks did not exist. The analysis of the cause-specific hazard models the event of interest in the absence of competing risk events and thus is the appropriate method.

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Reply Latouche *et al.*

> *Pintilie's work has the potential to further obscure the issues . . . Our main critique concerns the inaccurate assertion: "When modelling the cause specific hazard, one performs the analysis under the assumption that the competing risks do not exist".*

# Exercise I.5.4

Comment on the following definition of a competing risk:

> *A competing risk is an event whose occurrence either precludes the occurrence of the event under examination or fundamentally alters the probability of occurrence of that event.*

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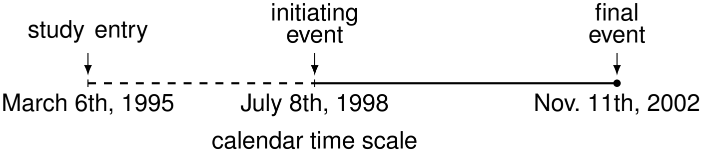
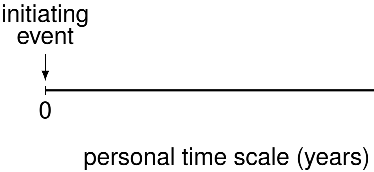
# Time scales

- Data collection in calendar time




- Often personal/patient time scale more relevant



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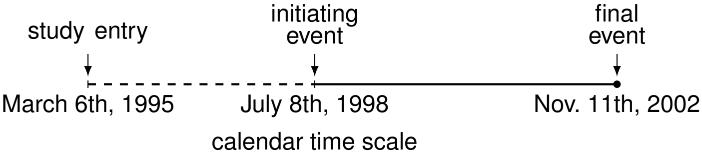


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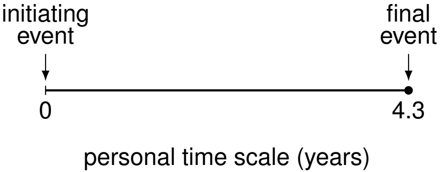


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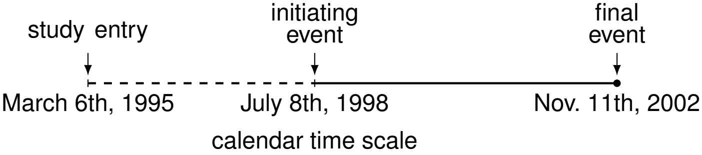


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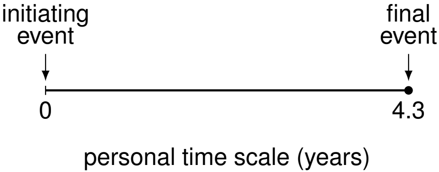
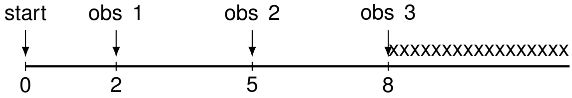



- Every time scale has a reference value: origin $\rightarrow$ end
- Classical survival analysis/competing risks: often single principal time scale

# Outline

# The observation process: censored event time

- Right censored: event free until last observation time




event happened after observation time 8

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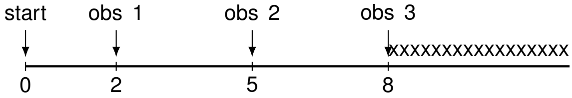


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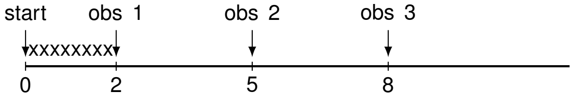
- Left censored: event before first observation time




event happened before observation time 2

# The observation process: censored event time

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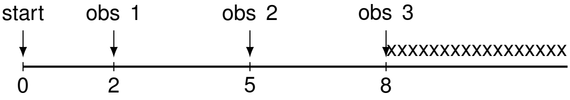


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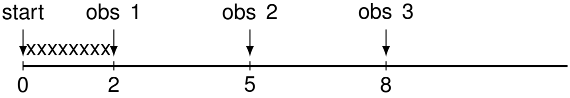
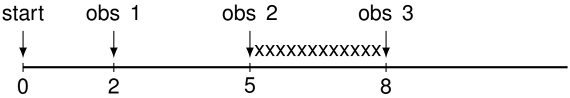



event happened before observation time 2

- Interval censored: event between two observation times




event happened between observation times 5 and 8

# Right censored event time: three reasons

Administrative censoring At date of analysis. Still in follow-up and event may be observed in the future

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**Administrative censoring** At date of analysis. Still in follow-up and event may be observed in the future

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**Competing risks** Two types:

- **Absorbing event:** other event (e.g. death) prevents occurrence of event of interest. No further follow-up information
- **Non-absorbing event:** "artificial censoring". Still in study, but conditions have changed too much to stay included.
  Another example: first event analysis

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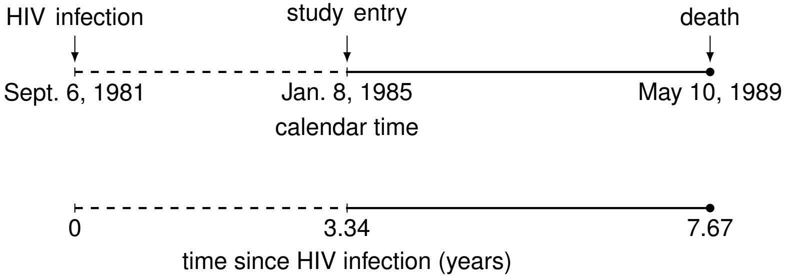
# Late entry

Individuals enter risk set after time origin



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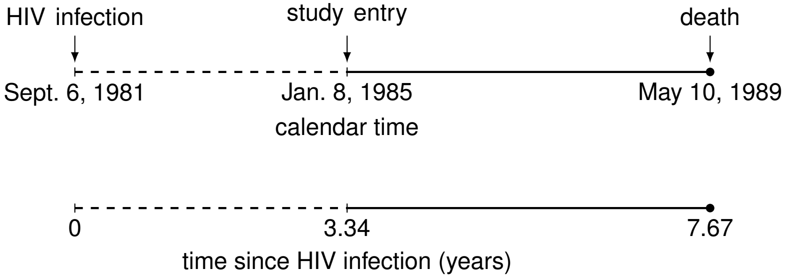
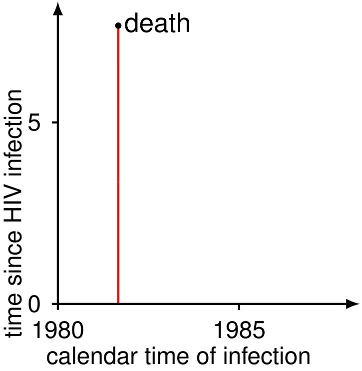



- Late entry usually induces length biased sampling (left truncation)

# Left truncation

Fast progressors underrepresented

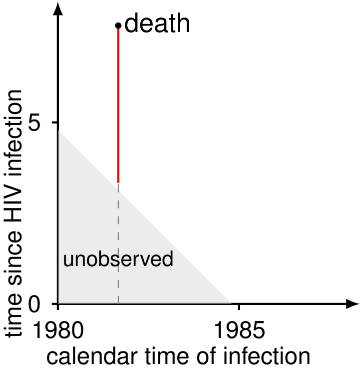
- Example: time from HIV infection to death



# Left truncation

Fast progressors underrepresented

- Example: time from HIV infection to death
- study started in 1985
- individual missed if died before 1985
- individual infected in 1982 only at risk from 1985



# Left truncation

## Fast progressors underrepresented

- Example: time from HIV infection to death
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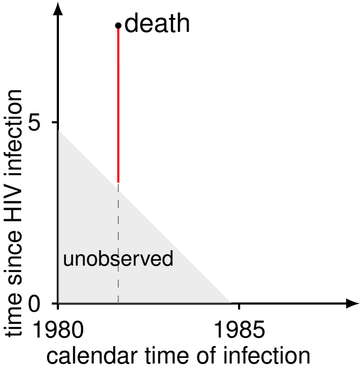
- individual missed if died before 1985
- individual infected in 1982 only at risk from 1985
- not same as left censoring



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- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \ldots, N$
- Assume $(V_i, C_i)$ independent of/non-informative for $(T_i, E_i)$ (possibly conditional on covariables in regression model)

# Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type $k$
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number "in risk set" (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \ldots, Z_{iq})$